SEMTM0051 Foundations of AI

SAMPLE PAPER

This is intended as an example of what an exam paper will look like, we have tried to make it similar in scope and difficulty, every time we set a sample paper students complain afterwards that the sample paper was misleading because it was easier than the real paper, but really, we've tried to make them the same level!

This paper contains two parts.

The first section contains 25 short questions.
Each question is worth two marks and all should be attempted.

The second section contains three long questions.
Each long question is worth 10 marks.
The best two long question answers will be used for assessment.

The maximum for this paper is 70 marks.

A sheet of formulae is provided at the end of the paper.

Other Instructions:
Only non-programmable calculators may be used.
Calculators must have the Faculty of Engineering Seal of Approval.

TURN OVER ONLY WHEN TOLD TO START WRITING

MODEL ANSWERS AND MARK SCHEME

Section A: short questions – answer all questions

Q1. A regression model is trained by least squares to predict y with a single constant c, that is, the model has no input. The training targets are $y = 2, 4, 6, 8, 10$. What value of c minimises the squared-error loss?

Answer

c is the mean, $c^* = 6$. [1 for identifying the mean, 1 for the value.]

Q2. In gradient descent the parameters are updated as $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta \nabla_{\boldsymbol{\theta}} L$. What goes wrong if the learning rate η is set too small, and what goes wrong if it is set too large?

Answer

Too small: each step is tiny, so training takes very many iterations to make progress. Too large: the update overshoots the minimum and the loss oscillates or diverges. [1 each.]

Q3. Explain how the model $y = a + b \sin x + c \cos x$ can be fitted using linear regression, even though it is not linear in x.

Answer

Use the feature vector $\phi(x) = (1, \sin x, \cos x)$; the model is linear in the parameters (a, b, c), so the normal equations or gradient descent for linear regression apply unchanged with $\mathbf{X}$ built from these features. [1 for features, 1 for "linear in parameters".]

Q4. What is 'softmax'?

Answer

Softmax turns a vector of real numbers into a probability distribution: $\mathrm{softmax}(\mathbf{a})_i = e^{a_i} / \sum_j e^{a_j}$, so the outputs are positive and sum to one, with the largest input getting the largest probability. [1 for "converts to probabilities", 1 for the formula.]

Q5. A three-class classifier outputs logits $(1, 1, 0)$. Compute the softmax probabilities to three significant figures; the softmax formula is on the formula sheet.

Answer

$e^1 = 2.718$, denominator $2.718 + 2.718 + 1 = 6.437$; probabilities $(0.422, 0.422, 0.155)$. [1 for the method, 1 for the numbers.]

Q6. Why is the output of a binary classifier passed through a sigmoid function before computing the loss?

Answer

The network output can be any real number, but the loss needs a probability; the sigmoid squashes the real line to $(0, 1)$ so the output can be interpreted as $P(y = 1 \mid \mathbf{x})$. [1 for "needs a probability", 1 for "sigmoid maps to $(0, 1)$".]

Q7. A binary classifier assigns probability 0.9 to the positive class for a particular input. Compute the cross-entropy loss for this input if the true label is positive, and if the true label is negative; the formula is on the formula sheet.

Answer

Positive: $-\ln 0.9 = 0.105$; negative: $-\ln 0.1 = 2.303$. [1 each.]

Q8. What do the extra layers of a deep neural network add that makes it better for complicated data?

Answer

Each extra layer computes features of the features from the layer below, so the network builds up a hierarchy of increasingly abstract representations; a complicated function can then be represented with far fewer units than a single very wide hidden layer would need. [1 for composition/hierarchy of features, 1 for efficiency compared with one wide layer.]

Q9. Sketch a sigmoid, $\sigma(x)$.

Answer

An S-shaped curve rising from 0 to 1, passing through $(0, \frac{1}{2})$. [2; 1 if the shape is right but the asymptotes are wrong.]

Q10. An affine layer takes an input row $\mathbf{x}$ with 50 coordinates and produces 10 outputs according to $\mathbf{a} = \mathbf{x}\mathbf{W} + \mathbf{b}$. How many trainable parameters does it have, including the biases?

Answer

$50 \times 10 + 10 = 510$. [1 for weights, 1 for remembering the bias.]

Q11. A network is built from two linear layers with no nonlinearity in between: $\mathbf{y} = \mathbf{x}\mathbf{W}_1\mathbf{W}_2$. Explain why this is no more expressive than a single linear layer. For simplicity biases are zero in this question.

Answer

$\mathbf{W}_1\mathbf{W}_2 = \mathbf{W}$ is itself a single matrix, so the composition is exactly a linear map $\mathbf{y} = \mathbf{x}\mathbf{W}$. Nonlinearities are what make depth useful. [2]

Q12. A neuron has preactivation $a(x) = 2x - 1$ and uses ReLU. Write the corresponding activation as a piecewise-defined function and sketch it.

Answer

$\phi(x) = 0$ for $x < \frac{1}{2}$ and $\phi(x) = 2x - 1$ for $x \geq \frac{1}{2}$; the sketch is flat at zero until $x = \frac{1}{2}$ and then a straight line of slope 2. [1 for the piecewise form with the correct breakpoint, 1 for the sketch.]

Q13. What is an epoch? A training set has $10\,000$ examples and the minibatch size is 100; how many parameter updates are made in one epoch?

Answer

One pass through the whole training set. $10\,000/100 = 100$ updates. [1 each.]

Q14. What is underfitting? Describe what you would see in the training and validation losses if a model is underfitting.

Answer

The model is not flexible enough, or not trained for long enough, to capture the structure in the data. Both training and validation losses stay high, and close to each other. [1 each.]

Q15. What is an autoencoder? Why do we use them?

Answer

A network trained to reproduce its input at its output, through a narrower intermediate layer. Because the input must pass through the bottleneck, that layer is forced to learn a compressed representation of the data, which can be used for compression, denoising, as features for another task, or as the basis of a generative model. [1 for what it is, 1 for why.]

Q16. How does a variational autoencoder differ from an autoencoder?

Answer

The encoder outputs the mean and variance of a distribution over the latent rather than a single code, and the value passed to the decoder is sampled from that distribution; the loss also includes a term pulling the distribution towards a prior, so that the latent space can be sampled from to generate new data. [1 for mean/variance and sampling, 1 for the prior/KL term.]

Q17. Describe how a new data point is generated from a trained variational autoencoder.

Answer

Sample a latent $\mathbf{z}$ from the prior $p(\mathbf{z})$, usually $\mathcal{N}(0, \mathbf{I})$, and pass it through the decoder to obtain $\mathbf{x}$. [1 each.]

Q18. A sparse autoencoder is trained with the loss $\|\mathbf{x}-\mathbf{W}_d\mathbf{h}\|^2+\lambda\|\mathbf{h}\|_1$ where $\mathbf{h}=\mathrm{ReLU}(\mathbf{W}_e\mathbf{x}+\mathbf{b})$ and the hidden dimension is much larger than the input dimension. What does the $\lambda\|\mathbf{h}\|_1$ term encourage, and why is the hidden layer chosen to be so wide?

Answer

The L_1 penalty pushes most hidden activations to exactly zero, so each input is explained by a few active features. The wide hidden layer gives a large dictionary of candidate features so that many distinct, individually interpretable features can be represented, each used rarely. [1 each.]

Q19. A 64×64 image is passed through a convolution with a 3×3 kernel, stride 1 and padding of one pixel on each side, followed by 2×2 max pooling. What are the spatial dimensions after each operation?

Answer

Convolution: 64×64, since the padding exactly compensates for the kernel; pooling: 32×32. [1 each.]

Q20. How many trainable parameters does a convolutional layer with eight input channels, 32 output channels and a 5×5 kernel have, including biases?

Answer

$32 \times 8 \times 5 \times 5 + 32 = 6400 + 32 = 6432$. [1 for weights, 1 for total with bias.]

Q21. A convolutional layer has far fewer parameters than a fully connected layer acting on the same image. Explain why.

Answer

The same small kernel is applied at every position, so the weights are shared across the image, and each output depends only on a small local patch rather than on every pixel. [1 for weight sharing, 1 for locality.]

Q22. In attention, each token produces a *query*, a *key* and a *value*. Using the analogy of looking something up in a dictionary or database, explain the role of each of the three.

Answer

The query is what the current token is looking for; keys are what each token advertises itself as, compared with the query to decide how much to attend; values are the content that is actually retrieved and mixed together according to the attention weights. [2 for all three; 1 for two.]

Q23. What is the difference between self-attention and cross-attention?

Answer

In self-attention the queries, keys and values all come from the same sequence; in cross-attention the queries come from one sequence and the keys and values from another, for example a decoder attending to an encoder's output. [1 each.]

Q24. What is a causal mask in self-attention, and why is it needed when training a model to predict the next token?

Answer

A mask that sets the score of position i attending to any position $j > i$ to $-\infty$, so each token only sees itself and earlier tokens. Without it the representation at position i could read the token at $i + 1$ directly, making next-token prediction trivial during training and impossible to reproduce at generation time. [1 each.]

Q25. What is multi-head attention, and why are several heads used rather than one?

Answer

Several attention operations run in parallel, each with its own query, key and value matrices, and their outputs are concatenated and projected. Different heads can attend to different things at once, for example one to the previous word and another to a matching bracket, which a single weighted average cannot do. [1 each.]

Section B: long questions – answer two questions

Q1. This question is about backpropagation. Consider the network

$$\mathbf{a}_1 = \mathbf{x}\mathbf{W}_1, \qquad \boldsymbol{\phi}_1 = \mathrm{ReLU}(\mathbf{a}_1), \qquad f = \boldsymbol{\phi}_1\mathbf{W}_2, \qquad L = (f-y)^2,$$

where f is short for $f_\theta(\mathbf{x})$ and with

$$\mathbf{x} = \begin{pmatrix} 2 & 1 \end{pmatrix}, \qquad \mathbf{W}_1 = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}, \qquad \mathbf{W}_2 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \qquad y = 1,$$

and all biases are fixed at zero.

(a) Compute the forward pass: $\mathbf{a}_1$, $\boldsymbol{\phi}_1$, f and L. **[3 marks]**

Answer

$\mathbf{a}_1 = \begin{pmatrix} 2\cdot 1 + 1\cdot(-1) & 2\cdot(-1) + 1\cdot 1 \end{pmatrix} = \begin{pmatrix} 1 & -1 \end{pmatrix}$; $\boldsymbol{\phi}_1 = \begin{pmatrix} 1 & 0 \end{pmatrix}$; $f = 1\cdot 2 + 0\cdot 1 = 2$; $L = (2-1)^2 = 1$. [1 for $\mathbf{a}_1$, 1 for $\boldsymbol{\phi}_1$, 1 for f and L.]

(b) Compute $\partial L/\partial f$ and hence $\partial L/\partial \mathbf{W}_2$. Note that $\partial L/\partial f = 2(f-y)$, and that f depends on the entry $W_{2,i}$ through the term $\phi_{1,i}W_{2,i}$. **[3 marks]**

Answer

$\partial L/\partial f = 2(f-y) = 2$ [1]. $\partial L/\partial W_{2,i} = 2(f-y)\,\phi_{1,i}$, so $\partial L/\partial \mathbf{W}_2 = \begin{pmatrix} 2 \\ 0 \end{pmatrix}$ [1 for the chain rule, 1 for the values].

(c) Compute $\partial L/\partial \boldsymbol{\phi}_1$ and then $\partial L/\partial \mathbf{a}_1$. Note that $\mathrm{ReLU}'(a) = 1$ if $a > 0$ and 0 when $a < 0$. **[2 marks]**

Answer

$\partial L/\partial \phi_{1,i} = 2(f-y)\,W_{2,i}$, so $\partial L/\partial \boldsymbol{\phi}_1 = \begin{pmatrix} 4 & 2 \end{pmatrix}$ [1]. $\partial L/\partial \mathbf{a}_1 = \begin{pmatrix} 4 & 2 \end{pmatrix} \odot \begin{pmatrix} 1 & 0 \end{pmatrix} = \begin{pmatrix} 4 & 0 \end{pmatrix}$ [1].

(d) Without doing any further calculation, state which column of $\mathbf{W}_1$ will not change in the gradient descent update, and explain why in one or two sentences. **[2 marks]**

Answer

The second column. The second hidden unit has $a_{1,2} < 0$, so its ReLU is inactive and $\partial L/\partial a_{1,2} = 0$; since $\partial L/\partial \mathbf{W}_1 = \mathbf{x}^\top\, \partial L/\partial \mathbf{a}_1$, whose jth column is $(\partial L/\partial a_{1,j})\,\mathbf{x}^\top$, the second column is zero. [1 for the column, 1 for the reason.]

Q2. This question is about self-attention in transformers. A sequence of three tokens has embeddings

$$\mathbf{x}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \qquad \mathbf{x}_2 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad \mathbf{x}_3 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

A single-head causal self-attention layer uses query, key and value matrices all equal to the identity, $\mathbf{W}_Q = \mathbf{W}_K = \mathbf{W}_V = \mathbf{I}$, so that $\mathbf{q}_i = \mathbf{k}_i = \mathbf{v}_i = \mathbf{x}_i$. Ignore the $1/\sqrt{d}$ scaling.

(a) The raw score for token i attending to token j is the dot product $\mathbf{q}_i^\top \mathbf{k}_j$. Write down the 3×3 matrix of raw scores. **[3 marks]**

Answer

$S_{ij} = \mathbf{x}_i^\top \mathbf{x}_j$: $\begin{pmatrix} 2 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$. [3; deduct 1 per wrong entry, minimum 0.]

(b) Which entries of this matrix does the causal mask remove, and why? **[2 marks]**

Answer

The entries above the diagonal, S_{12}, S_{13}, S_{23}, set to $-\infty$ so their softmax weight is zero, because token i must not attend to later tokens $j > i$ that it will not have seen at generation time. [1 each.]

(c) Compute the output of the attention layer at positions 1 and 2, to two decimal places. Position 1 can attend only to itself; position 2 has a two-way softmax over the scores S_{21} and S_{22}, and the output is the weighted sum of the values. **[3 marks]**

Answer

Position 1: weight 1 on itself, output $\mathbf{v}_1 = (1, 1)$ [1]. Position 2: scores $(1, 1)$, weights $(0.5, 0.5)$ [1], output $0.5(1, 1) + 0.5(1, 0) = (1, 0.5)$ [1].

(d) For a sequence of n tokens, how does the cost of computing the attention scores scale with n? Explain briefly why this is a problem for long sequences. **[2 marks]**

Answer

Every token attends to every other, so there are n^2 scores and the cost in time and memory is $O(n^2)$ [1]. Doubling the context length quadruples the cost, so very long documents become prohibitively expensive, which is why context lengths are limited [1].

Q3. This question is about probabilistic and sparse autoencoders. A variational autoencoder with encoder $q_\phi(\mathbf{z} \mid \mathbf{x})$, decoder $p_\theta(\mathbf{x} \mid \mathbf{z})$ and prior $p(\mathbf{z})$ is trained by maximising

$$\mathcal{L}(\theta, \phi) = \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}\big[\ln p_\theta(\mathbf{x} \mid \mathbf{z})\big] - \mathrm{KL}\big(q_\phi(\mathbf{z} \mid \mathbf{x}) \,\|\, p(\mathbf{z})\big).$$

(a) Explain in words what each of the two terms rewards or penalises, and what would go wrong if the second term were left out. **[3 marks]**

Answer

First term: the expected log-likelihood of the data under the decoder, that is, a reconstruction term rewarding codes from which $\mathbf{x}$ can be regenerated [1]. Second term: penalises the encoder's distribution for being far from the prior, keeping the latent space smooth and close to $p(\mathbf{z})$ so that it can be sampled from to generate new data [1]. Without it the encoder can make $\sigma \to 0$ and spread the means arbitrarily, collapsing to an ordinary autoencoder whose latent space has gaps and is useless for generation [1].

(b) For a one-dimensional latent with $q(z \mid \mathbf{x}) = \mathcal{N}(\mu, \sigma^2)$ and $p(z) = \mathcal{N}(0, 1)$, the KL term is $\frac{1}{2}(\mu^2 + \sigma^2 - 1 - \ln \sigma^2)$. Compute it for $\mu = 1$, $\sigma = 2$, to two decimal places. What values of μ and σ make it zero? **[2 marks]**

Answer

$\frac{1}{2}(1 + 4 - 1 - \ln 4) = \frac{1}{2}(4 - 1.386) = 1.31$ [1]. It is zero when $\mu = 0$ and $\sigma = 1$, that is, when q equals the prior [1].

(c) Evaluating the first term requires drawing a sample $z \sim \mathcal{N}(\mu, \sigma^2)$. Explain why backpropagation cannot pass through a random sampling step to reach μ and σ, and how writing $z = \mu + \sigma\epsilon$ with $\epsilon \sim \mathcal{N}(0, 1)$, the reparameterisation trick, solves the problem. **[3 marks]**

Answer

A sampler is not a differentiable function of its parameters: there is no $\partial z/\partial \mu$ for "z is a random draw", so the gradient chain is broken and the encoder gets no signal from the reconstruction term [1]. Writing $z = \mu + \sigma\epsilon$ puts all the randomness into ϵ, which does not depend on the encoder parameters; z is now an ordinary deterministic function of μ and σ, with $\partial z/\partial \mu = 1$ and $\partial z/\partial \sigma = \epsilon$, so backprop works as usual for a sampled ϵ [2].

(d) A sparse autoencoder is trained on the hidden activations of a language model. Explain what a "feature" of the sparse autoencoder is, and describe one way you could check whether a given feature is interpretable. **[2 marks]**

Answer

A feature is one hidden unit of the sparse autoencoder, equivalently one column of the decoder, a direction in the language model's activation space that is switched on for a small fraction of inputs [1]. To check it: collect the text inputs on which the feature is most strongly active and see whether they share an obvious property, such as all being about a particular topic or in a particular language; alternatively, add or remove the feature's direction from the activations and see whether the model's output changes in the expected way [1].

Formulae

Linear regression. Optimal parameters: $\boldsymbol{\theta}^* = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$, provided $\mathbf{X}^\top \mathbf{X}$ is invertible. Here $\boldsymbol{\theta}$ is the complete parameter vector, including the intercept, and when an intercept is fitted the design matrix $\mathbf{X}$ includes a column of ones.

Matrix inverse. $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \dfrac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$, assuming $ad - bc \neq 0$.

Sigmoid and softmax. $\sigma(x) = \dfrac{1}{1 + e^{-x}}, \qquad \operatorname{softmax}(\mathbf{a})_i = \dfrac{e^{a_i}}{\sum_j e^{a_j}}.$

Cross-entropy loss. Binary, with p the predicted probability of the positive class and $y \in \{0, 1\}$: $L = -y \ln p - (1 - y) \ln(1 - p)$. Multi-class with true class c: $L = -\ln \operatorname{softmax}(\mathbf{a})_c$.

Derivatives. $\sigma'(x) = \sigma(x)(1 - \sigma(x))$. $\mathrm{ReLU}'(a) = 1$ if $a > 0$ and zero when $a < 0$; often software will define the derivative as zero for $a = 0$ as well, but this is just a convention and not mathematically coherent.

2D convolution. $a_{x,y} = \displaystyle\sum_{\delta_x=-1}^{1} \sum_{\delta_y=-1}^{1} h_{x+\delta_x,\, y+\delta_y}\, W_{\delta_x, \delta_y}.$

Momentum. $\mathbf{v} \leftarrow \beta \mathbf{v} + \mathbf{g}, \quad \mathbf{w} \leftarrow \mathbf{w} - \eta \mathbf{v}.$

Adam. $\mathbf{m} \leftarrow \beta_1 \mathbf{m} + (1 - \beta_1)\mathbf{g}, \quad \mathbf{s} \leftarrow \beta_2 \mathbf{s} + (1 - \beta_2)\mathbf{g}^2, \quad \mathbf{w} \leftarrow \mathbf{w} - \eta \dfrac{\mathbf{m}}{\sqrt{\mathbf{s}} + \epsilon}$ (bias corrections omitted).

Scaled dot-product attention. $\mathrm{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \operatorname{softmax}\!\left(\dfrac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d}}\right)\mathbf{V}$, with softmax applied row-wise.

KL divergence between Gaussians. $\mathrm{KL}\big(\mathcal{N}(\mu, \sigma^2) \,\|\, \mathcal{N}(0, 1)\big) = \tfrac{1}{2}\left(\mu^2 + \sigma^2 - 1 - \ln \sigma^2\right).$

Evidence lower bound. $\ln p_\theta(\mathbf{x}) \;\geq\; \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}\big[\ln p_\theta(\mathbf{x} \mid \mathbf{z})\big] - \mathrm{KL}\big(q_\phi(\mathbf{z} \mid \mathbf{x}) \,\|\, p(\mathbf{z})\big).$

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